Applied Energy 323 (2022) 119560

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

Agrivoltaics analysis in a techno-economic framework: Understanding why agrivoltaics on rice will always be profitable

M. Sojib Ahmed [a], M. Rezwan Khan [b], Anisul Haque [a], M. Ryyan Khan [a,*]

[a] Department of Electrical and Electronic Engineering, East West University, Dhaka, Bangladesh
[b] Department of Electrical and Electronic Engineering, United International University, Dhaka, Bangladesh

HIGHLIGHTS

- Techno-economic model and analysis of bifacial agrivoltaics (AV) with paddy rice.
- Performance indicator of AV is economics driven—not land equivalent ratio (LER)
- Horizontal E/W AV layout has more economic gain than tilted N/S and vertical E/W.
- Net profit from AVs will be 22–115 times higher than conventional rice cultivation.
- AV with bifacial vs. monofacial panels can boost profit by around 18 to 35%

ARTICLE INFO

Keywords:
Agrivoltaics
Panel array
Bifacial
Energy yield
Paddy rice
Crop yield
Economics
Profit

ABSTRACT

Agrivoltaics (AV) promises sustainable food and energy production from shared lands. While various literature studies AVs for leafy crops, there are limited equivalent works on AVs with major crops (e.g., rice, corn, wheat, etc.). Here, we present an end-to-end modeling framework to analyze location-specific AVs over paddy rice. We consider the local ambient conditions to find the spatially distributed rice production under panel arrays, the panel yield, and the economic aspects (costs, revenue, and profit) of the AV system. On a given land, solar energy has a far larger absolute profit than rice cultivation due to the 100s of folds higher investment possible in panels compared to paddy rice. Policy-level constraints are required to set a permissible reduction in rice for more energy. We find that, while maintaining 90% (or 80%) of the conventional rice production, the overall profit can be 22–115 (or 30–132) times higher than rice cultivation.

1. Introduction and background

In the near future, meeting food and energy demand from existing energy resources and land will be a challenging issue. Although a substantial portion of global energy comes from fossil fuel-based energy sources [1], they have a significant environmental impact. Renewable energy sources, particularly solar energy, appear to be a promising solution to harness energy with minimal environmental impact. Additionally, the cost of solar energy has dropped significantly in the past couple of years due to a combination of reduced manufacturing costs and technological performance improvements of solar cells like bifaciality and tandem. Among those, bifacial technology is becoming more popular because these types of solar cells can produce more energy from the same footprint by capturing light on both the front and rear sides. This makes the energy cheaper than the conventional mono facial modules [2]. Bifacial module prices are expected to reduce over the years with increased production – the share of bifacial modules will grow by around 55% within 2031 [3]. Future solar energy expansion is therefore expected to be through bifacial panel arrays.

Besides the growing energy consumption, global food requirements will also increase by around 70% of current food production by 2050 [4]. These statistical analyses indicate future land use conflicts for energy and food production. That is why proper utilization of existing land and resources is an issue that we need to think about to keep balance and secure food and energy independence. One possible way could be to develop agrivoltaics (AV) technology.

Studies on microclimate and vegetable growth: The idea of AV is to co-utilize the resources of a single land by placing solar panel arrays on

* Corresponding author.
E-mail addresses: ryyan@ewubd.edu, ryyan.khan.eee@gmail.com (M. Ryyan Khan).

https://doi.org/10.1016/j.apenergy.2022.119560
Received 17 February 2022; Received in revised form 19 May 2022; Accepted 25 June 2022
Available online 10 July 2022

top of crops to enhance the net land productivity [5,6]. Some research groups have investigated the relative changes in crop development under panel arrays and their effects on land productivity [7–12]. For example, Marrou et al. [7] experimentally analyzed how microclimate conditions and crops (lettuce, cucumbers, and wheat) development differ for 50% and 70% radiation under photovoltaic (PV) array. They observed lower leaf growth under shading. Additionally, they found that soil temperature drops by around 2°C due to radiation fluctuation under the PV array. Another experiment by Marrou et al. [8] has shown the relative change in lettuce yield under 70% and 50% fractional radiation. They did not find any significant change in radiation conversion efficiency (or lettuce yield). Adeh et al. [9] have found around 90% more biomass (grass) as well as 328% water efficiency compared to the control condition. Barron-Gafford et al. [10] have demonstrated that an AV setup ensures proper balance among PV energy, crop (Chiltepin pepper, Jalapeno, and Tomato) production, and water usage efficiency by decreasing temperature and increasing soil moisture content. They observed double crop production from AV setup compared to conventional farming. Also, they found 3% more energy during crop growing seasons due to temperature reductions on the PV panel surface. Weselek et al. [11] performed an experiment to investigate the potentiality of cultivating celeriac under solar panels. They found that the presence of solar PV reduced photosynthetic active radiation (PAR) by around 30%, although there was no significant change in celeriac yield. A numerical study by Valle et al. [12] showed how tracking AV system design (such as sun-tracking and control tracking) and stationary AV configuration (considering 70% of transmitted light on crop) affects microclimatic variables, lettuce growth, and PV performance. They found higher land equivalent ratio (LER) from the sun tracking AV system throughout all seasons (around 1.5) due to the higher energy contributions of this system. The control tracking configuration provides lower LER (around 1.25) while ensuring the best microclimate for crops. Panel output and crop yield are highly dependent on the local ambient conditions (e.g., radiation, air temperature, soil temperature). A comprehensive location and ambient condition aware study are required to assess the potential land-productivity gain of a well-designed AV.

AV on Major crops: While several studies in the literature consider AV system experiments and analysis with leafy crops and vegetables, there are very limited equivalent works on AVs with major crops (e.g., rice, wheat, corn, maize, etc.). The radiation and temperature requirements for the development of major crops are different from the vegetables [13–15]. Gonocruz et al. [14] experimentally showed that rice yield linearly falls with increasing amounts of partial shading. Numerical modeling for vertically mounted panel array over rice-field by Vijayan et al. [16] shows nearly 54% extra land productivity (at the cost of reduced rice yield) in Chennai, India. Dupraz et al. [17] found around 37% and 73% extra land productivity based on wheat yield for 50% and 70% radiation under photovoltaic (PV) arrays, respectively.

Although the studies discussed above presented higher combined benefits from single land use, the optimized performance of an AV system would be found based on crop category and PV system design parameters [18–21]. Recently, Trommsdorff et al. [18] showed the effect of AV design on light availability on crops. They found a 78% extra land productivity for AV with wheat. However, their model did not consider the effect of temperature on wheat yield. Riaz et al. [19] showed the optimized performance variability between two AV configurations– East-West (E/W) facing vertical bifacial and tilted North-South (N/S) orientated monofacial. Another study [20] by the same group showed crop-specific optimum light productivity outcomes as Light Productivity Factor (LPF) for several AV farm settings, including tracking. They found LPF 1.2 to 2 based on crop's shade tolerance capacity. However, they used photosynthetic active radiation (PAR) as a measure of crop production instead of the weather variable crop model. Also, location-specific variability has not been explored there. For oats and potatoes, Campana et al. [21] developed an optimum vertical AV system for Västmanland (Sweden) and found the LER to be over 1.2. Although the vertical AV arrangement has some additional advantages, it contributes less energy, which ultimately affects total land productivity [19]. Even though the combined land productivity has been shown to increase in all the experimental and numerical studies, major-crop production always degraded.

Economic gain: Most prior research considered the land equivalent ratio (LER) as a potential indicator. However, just as any PV system or crop farm, the viability of a system will be decided by its overall economic gain. The financials of panel arrays will not be the same in solar farms versus in AVs. The Levelized Cost of Energy (LCOE) of AV can be roughly 38% higher than traditional ground-mounted PV [22]. A recent study by Thompson et al. [23] found 2.5% and 35% economical gain for basil and spinach, respectively in Italy. Dinesh and Pearce et al. [24] have found 30% more economic benefits from AV than the conventional farming for lettuce in the USA. Using an analytical framework, Feuerbacher et al. [25] predicted that the cereal and vegetable farming systems in Southern Germany could adopt agrivoltaics at tariffs of 8.63 and 9.00 EUR-cents per kWh, respectively. Each of these studies has shown the gross economic gain for a particular location — it is unclear if the predictions hold across the borders.

Gaps in literature: Prior literature indicates that vegetable and leafy crop development under PV arrays has been better studied compared to major crops. Since a large portion of global agricultural land is currently being used to cultivate the major crop, incorporating PV modules onto those fields shows prospects for large-scale AV expansion. Additionally, while LER has been used as a key performance indicator of AVs in many prior works, the LER may not be the appropriate metric for evaluating the optimal design of an AV system [21,26]. The LER does not account for the proper weightage of energy and crop prices. From beginning to end, we need to consider the economic aspect of an AV system. The economic gain or profit of a given crop across the globe has not been studied in the literature.

This work: In this paper, our main target is to investigate the location-specific bifacial AV performance for rice (*Oryza sativa*) and its economic viability. We chose six rice-producing locations- An Giang (Vietnam) ($10.5^{\circ}N, 105.12^{\circ}E$), Dhaka (Bangladesh) ($23.8^{\circ}N, 90.45^{\circ}E$), Jiangsu (China) ($33.2^{\circ}N, 119.78^{\circ}E$), Damietta (Egypt) ($31.1^{\circ}N, 31.8^{\circ}E$), Rio Grande do Sul (Brazil) ($9.1^{\circ}S, 36.95^{\circ}W$), and Haryana (India) ($29.05^{\circ}N, 76.2^{\circ}E$). The countries listed above contribute significantly to global rice production [27]. We explore the following key questions: (i) How do the location-specific weather conditions affect the performance of AV systems? (ii) What are the effects of optimal N/S, vertical E/W, and horizontal E/W AV arrangements in terms of rice yield and PV energy? (iii) For these AV systems, how do rice yield and PV energy output vary with panel array row spacing? (iv) How do local financial parameters affect the economic output of these AV farm configurations? and (v) Is there any additional economic advantage of bifacial AV over monofacial? To answer these, we have created an end-to-end location-specific AV modeling framework. For a given location and bi/monofacial panel array configuration (e.g., orientation, pitch, tilt angle, fixture height), we find the time-varying irradiance on each point on the cropland and panel surface. The local temporal weather (temperature and precipitation) along with the time-varying spatial distribution of sunlight on the crop land is coupled to the Agricultural Production System Simulator (APSIM) [28,29]. This allows us to find spatially distributed rice production. The panel array yield is calculated from the irradiance and electrical model. We analyze the system potentiality for three different AV configurations (tilted N/S, vertical E/W, and horizontal E/W). Since higher PV density penalizes the crop output of an AV farm, we do a parametric analysis to find optimal row spacing to ensure the desired amount of rice production per unit of land area. The AV configuration and the yield are simultaneously considered in the economic model (initial cost, maintenance, and output) to estimate the revenue and profit. The economics is country specific as local costs are considered. The economic output and rice production requirement will

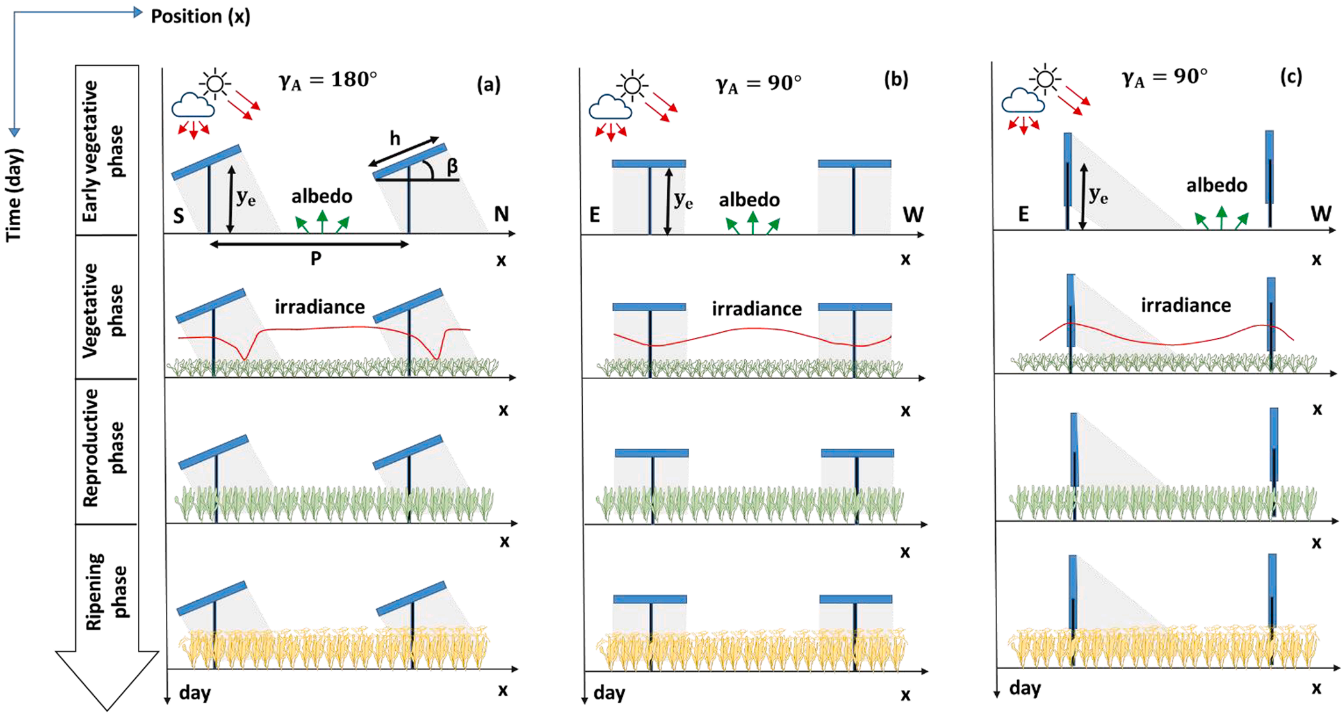

Fig. 1. Schematic of bifacial AV farm (a) tilted N/S (b) horizontal E/W (c) vertical E/W. The solid red line represents the daily average irradiance on each point on the cropland. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

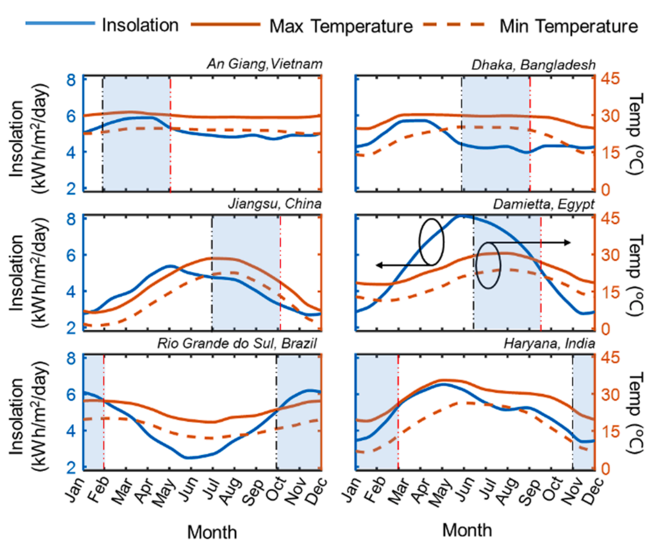

Fig. 2. Monthly insolation (Blue line and left y-axis), Maximum temperature (Red solid and right y-axis), and Minimum temperature (Red dash and right y-axis) are shown. Shaded regions represent the rice cultivation season for each location. The black and red long dash-dot lines indicate the beginning and end of the rice season [31]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

dictate the PV density optimization. Our end-to-end analysis shows that horizontal E/W AV layout is the best option economically, irrespective of the location. In general, the investment per land area for solar panel arrays is 100s of times higher than that possible in rice cultivation. The net profit is therefore much higher from the PV system compared to rice (even though rice production has a higher profit per investment). Hence, there is no optimum array pitch for AV systems, and a PV-only system will always bring in higher absolute profit, even though the high panel density reduces crop production under shading. That is why a minimum rice-production requirement should be set at a policy level if croplands are to be utilized for AV. For example, if rice production in AV is set to 80% of that compared to conventional open-field agriculture, the profit increases by more than a factor of 30 for most locations in our study.

2. Numerical model

Fig. 1 represents the typical scenario of agrivoltaics farms. We consider infinitely long rows and periodic PV array setup. This allows us to consider the PV farm modeling framework in 2D. A single period analysis will be enough due to the periodicity. The edge effects of PV array may be considered for finite-size PV farms—this effect is, however, small for large area systems. For modeling an AV farm, several design parameters need to be considered such as: the tilt angle of panel β, row spacing or pitch p, panel elevation from the ground y_e, panel width h, and array azimuth angle γ_A, albedo R_A. Location-specific ambient conditions (e.g., insolation, temperature) and economic factors (e.g., costs, selling price) along with the design parameters determine the final yield and profit. For a given location, the amount of incident light on the PV and the crop surface will be determined by the design parameters and ambient conditions. Light will have a non-uniform distribution over the cropland due to shadows (direct and diffuse) cast by the solar panels. A fraction of the incoming sunlight will reflect off the crops onto the panel surfaces. We need a crop model to find rice yield distributions over the ground in the presence of PV which will account for the local ambient conditions, including temperature, radiation, and precipitation. All these effects in the AV system are integrated within a modeling framework through multiple sub-models: (i) location-aware irradiance model, (ii) light collection (on panel and ground) model, (iii) temperature-dependent panel efficiency model, (iv) PV output electrical model, and (v) rice yield model. Finally, an economic model is used to analyze the revenue of the entire AV system based on location-specific initial and maintenance costs and selling prices. The final result is, of course, dependent on the system design criteria described before. As a result, the system can be optimized under prescribed economic or yield constraints. We have gone through each of the steps in detail in the ‘modeling methods’ section (Appendix A).

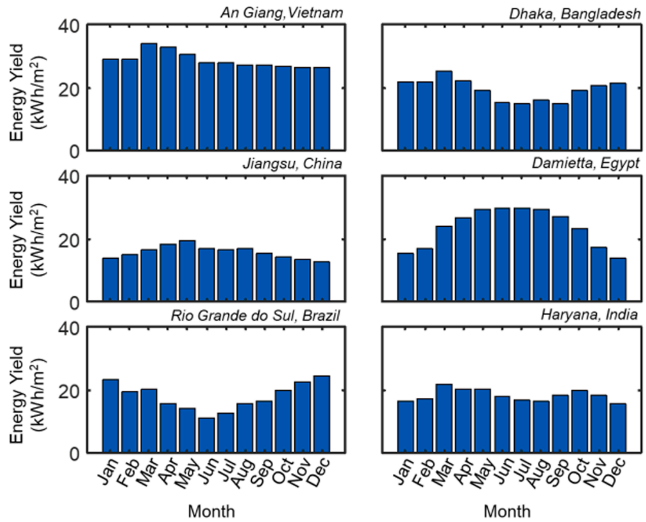

Fig. 3. Monthly energy yield for the optimized south-facing bifacial farm. The yearly energy yield (*YEY*) for a location can be found by summing the monthly outputs.

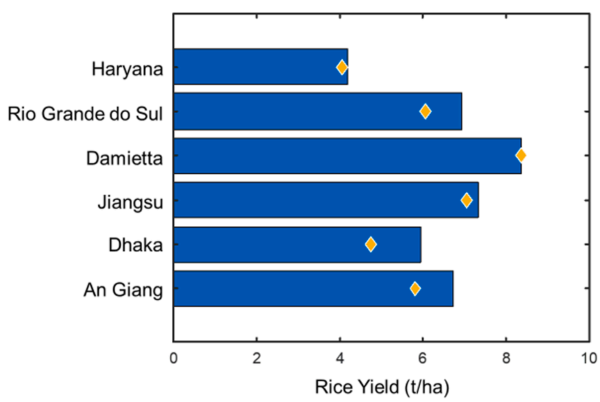

Fig. 4. Paddy rice Yield for conventional farming conditions. Diamond symbols represent the historical paddy production data [37].

3. Conventional farm output

In this section, we will discuss the baselines for the energy production of a conventional solar farm and rice yield in open field-condition (no shading). We have selected six rice-producing locations for this study:

1. An Giang, Vietnam (10.5°N,105.12°E),
2. Dhaka, Bangladesh (23.8°N,90.45°E),
3. Jiangsu, China (33.2°N,119.78°E),
4. Damietta, Egypt (31.1°N,31.8°E),
5. Rio Grande do sul, Brazil (9.1°S,36.95°W), and
6. Haryana, India (29.05°N,76.2°E).

Ambient environmental conditions for these locations are shown in Fig. 2 [30]. Rice cultivation seasons vary from location to location – the relevant season for each location has been marked by the shaded regions in Fig. 2. Crop seasons are chosen from the crop calendar of the above-mentioned countries [31]. For our estimation of two-season rice production, after a single season calculation, we assume that profit from the rice will approximately double – we will further discuss this after the results. Also, crop rotation is very common in some cases – we do not consider this in our present analysis.

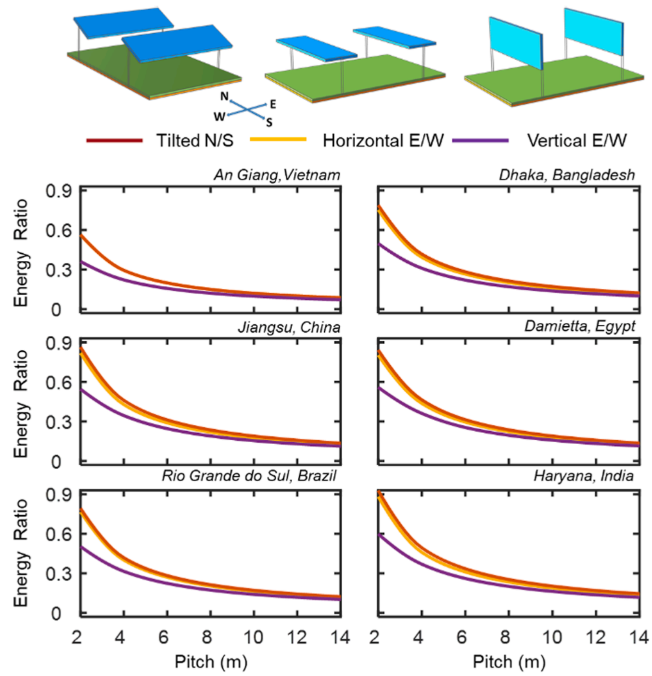

Fig. 5. The energy yield contributions of several agrivoltaics farm setups, including tilted N/S (Red), h-E/W (Orange), and v-E/W (Purple) are shown as a function of array pitch. In all cases, tilted bifacial N/S performs best. The vertical bifacial arrangement produces less energy than the other two. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.1. Solar farm output: Optimized N/S farm

There are several design considerations (e.g., pitch, tilt angle, panel elevation from the ground) to obtain electricity more cost-effectively from solar farms. Typically, North or South (N/S) oriented bifacial farms are optimal [32–34]. In our analysis, for the baseline N/S bifacial farms, we use the location-specific optimal configurations found in ref. [34]. Fig. 3 shows the monthly energy output (per land area) for different locations. Except for March-April, the energy production in An Giang (Vietnam) has remained relatively constant. The generation of energy for Haryana (India) and Dhaka (Bangladesh) follows a similar trend. Being in the southern hemisphere, Rio Grande do Sul (Brazil) has an energy peak around November-March. Damietta (Egypt) and Jiangsu (China) have energy peaks around May to August and April to September, respectively. The yearly energy yield (*YEY*) can be found by summing the monthly yields for respective locations.

3.2. Paddy rice output: Open farm condition

Fig. 4 shows the open farm rice yield output for different locations, calculated as described in the rice yield modeling section (see Modeling Methods section). We use the default soil characterization data (set for South Asia) from APSIM for all location except Haryana. The soil information from ref. [35,36] are used for Haryana for better predictions. Diamond symbols represent the country-wise average rice yield data [37]. For all locations, the practical rice yield data match closely with that predicted by our model using APSIM. The GHI contribution in Damietta is high, so it is expected that rice yield will be high at this location.

4. Location-specific agrivoltaics farm analysis: Agrivoltaics farm vs. conventional farm

In this section, we will compare the location-specific performance

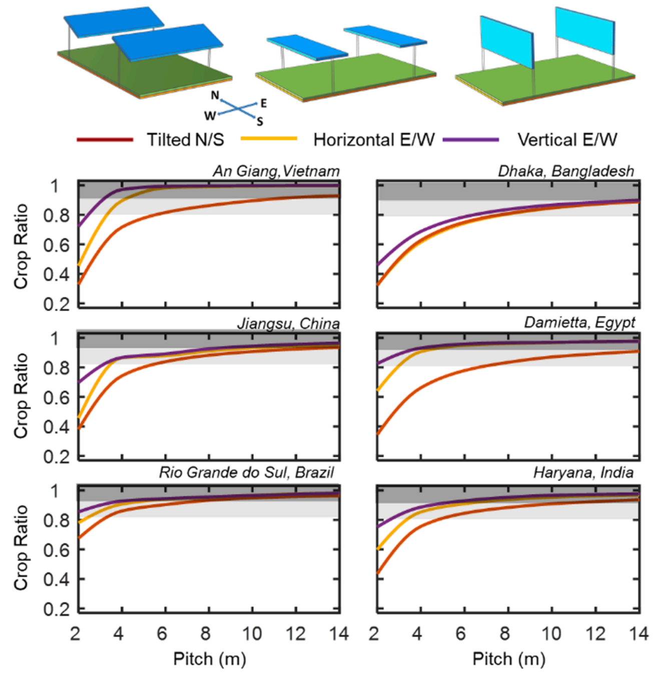

Fig. 6. The crop yield (paddy rice) from several agrivoltaics farm setups, including tilted N/S (Red), h-E/W (Orange), and v-E/W (Purple) are shown as a function of array pitch. The deep and light gray shaded regions indicate the traditional (unshaded) farm's 90% and 80% crop yield, respectively. For vertical bifacial arrangement, rice production becomes higher than in the other two configurations due to the lower ground coverage of the panels. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(energy and rice yield production) of three different AV farm configurations:

(i) North-South (N/S) AV: The panel array faces the north or south ($\gamma_A = 0^{\circ}$ or 180°) in the southern or northern hemisphere. The panels are tilted optimally according to their location [34].
(ii) Vertical East-West (v-E/W) AV: Vertical panel arrays face east/west ($\gamma_A = 90^{\circ}$, $\beta = 90^{\circ}$).
(iii) Horizontal East-West (h-E/W) AV: This is similar to v-E/W except that the panels are kept horizontal ($\gamma_A = 90^{\circ}$, $\beta = 0^{\circ}$).

We will discuss the rice yield distribution over the ground and the panel array output for different panel densities. For all our calculations, we assume panel width $h = 1$ m and ground to panel elevation $y_e = 3$ m.

4.1. Power output for different agrivoltaics configurations: Agrivoltaics farm vs. conventional farm

We will discuss the bifacial panel output (per land area) in AV setup in terms of 'energy ratio', defined as the ratio of AV energy output to optimum N/S PV farm output (see Fig. 3 for optimum N/S PV farm yield). Fig. 5 shows the energy ratio for different AV farm setups with varying pitches. The tilted-N/S configuration yield is always higher than the rest of the two configurations for all locations. However, the energy ratio of tilted N/S and h-E/W AV configurations are very close. At lower pitch (e.g., $p = 2m$ or $p/h = 2$), tilted N/S and h-E/W clearly outperform v-E/W bifacial AV farms. Mutual shading between adjacent vertical panels in v-E/W AV significantly reduces the energy yield at lower pitches.

We can intuitively guess that the low shadow constraint for rice production will require a higher panel array pitch. In such cases, the albedo and diffuse light contribution can significantly increase, justifying the importance of using bifacial panels. In the next section, we will discuss the crop production distribution over the ground and the change in crop yield due to varying panel density.

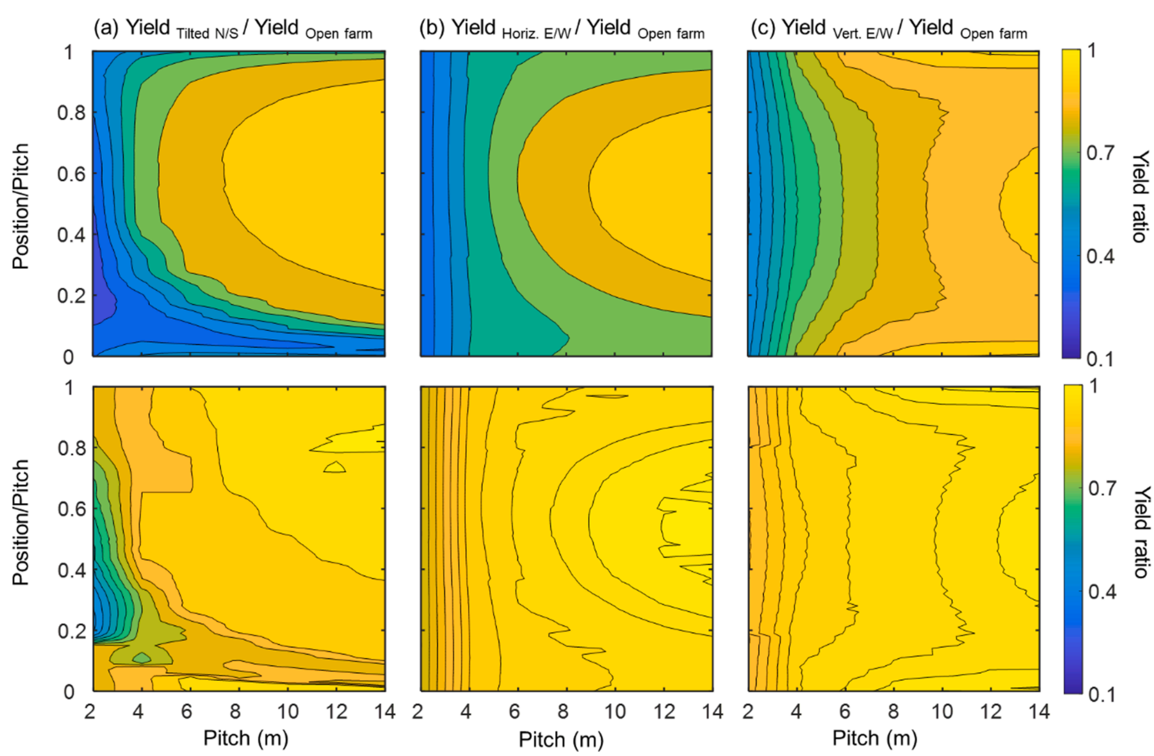

Fig. 7. Crop yield distribution over the ground compared to conventional farm output as a function of the pitch for tilted N/S (left column), horizontal E/W (middle column), and vertical E/W (right column) agrivoltaics farm at Dhaka (top row) and Rio Grande do Sul (bottom row).

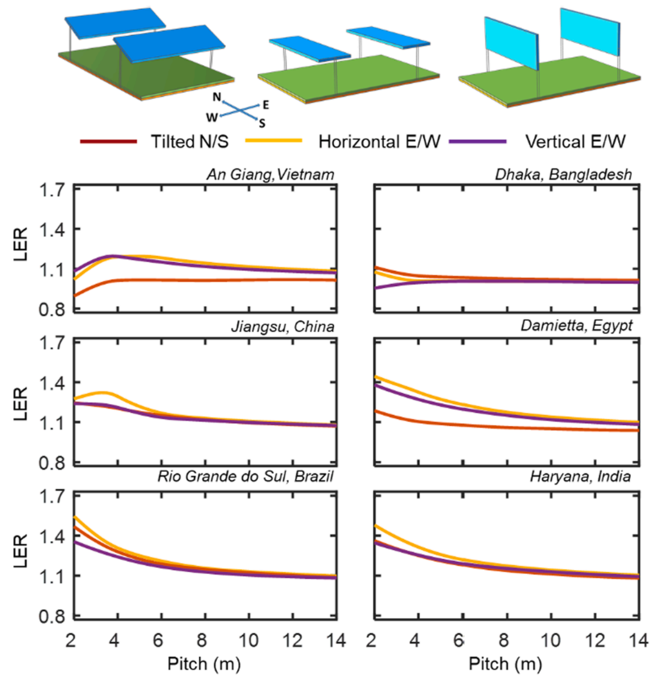

Fig. 8. Land Equivalent Ratio (LER) as a function of the pitch for various PV configurations, namely tilted N/S (Red), horizontal E/W (Orange), and vertical E/W (Purple) for different locations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.2. *Crop output for different agrivoltaics configurations: Agrivoltaics farm vs. conventional farm*

For various AV farm setups, Fig. 6 indicates the paddy production compared to conventional agricultural farms (see Fig. 4) as a function of pitch. The deep and lighter gray shaded regions indicate the traditional farm's 90% and 80% crop yield, respectively. Crop production increases with the increasing row distance in all locations and saturates at higher pitch values. Most locations, except for Dhaka and Haryana, can reach 90% crop production at $p = 4m$ or $p/h = 4$. For the Rio Grande, both E/W oriented setups (v-E/W and h-E/W) can reach 80% crop ratio at $p = 2m$ or $p/h = 2$. For all locations, the AV farm setup with v-E/W and h-E/W have better crop production than the N/S setup. For E/W configurations, the shadow moves on the crop surface with varying times. On the other hand, the shadow remains fixed under the N/S facing panel. So crop yield under the N/S oriented panel affects more than the other two compartments. A highly elevated structure and sufficient space between mutual rows are required for h-E/W and tilted N/S AV arrangements to operate farming equipment for crop cultivation and harvesting [24]. This will increase the installation cost of the AV farm. The vertical bifacial may provide a better logistical choice – however, we will not separately analyze such scenarios.

4.3. *Crop production distribution over the ground*

Fig. 7 shows the crop yield distribution over the ground between mutual rows compared to open farm conditions for three AV farm setups in Dhaka (Bangladesh) and Rio Grande do Sul (Brazil). The y-axis represents the position on the ground normalized to the pitch (i.e., $x_n = x/p$) – this way x_n always ranges between [0, 1] for any pitch p. The crop yield distributions over the ground for the rest of the locations are shown in Fig. S1 (supp. doc.). At higher panel density (i.e., lower pitch), crop yield is minimum for all configurations due to a higher fraction of shadow coverage.

Due to the sun path, there is a shadow close to the panels for most of the time of the day. The constant shadow under the N/S orientated panel severely reduces crop production. It may be possible to cultivate a shade-tolerant crop instead of paddy rice under the N/S oriented PV row as a more economically viable option. On the other hand, for E/W configurations, the shadow continuously moves over the crop surface throughout the day – this gives a more uniform spatial distribution of radiation and crop production.

We found that crop production is relatively higher towards the middle ($x_n = 0.5$) of two rows for the h-E/W setup. Also, crop production directly under the vertically oriented panel is higher than the other two PV configurations due to lower shading in the v-E/W PV arrangement. Interestingly, the constant shadow under the N/S configured panel is less pronounced for Rio Grande do Sul compared to the other locations in this study. This difference arises due to the crop seasons and the corresponding sun path variation.

5. Agrivoltaics farm's output

In this section, we will discuss the combined production in AV as well as the economic output variability as a function of pitch. We will also discuss the optimized performance and the corresponding profit under certain crop production requirements.

5.1. *Land equivalent ratio (LER)*

The Land Equivalent Ratio (LER) is widely used as an AV performance indicator in literature. It is defined as follows:

$$LER = \frac{YEY_{APV}}{YEY_{Open}} + \frac{YCY_{APV}}{YCY_{Open}}$$
$$= (\text{Energy Ratio}) + (\text{Crop Ratio})$$

LER $= 1$ indicates that the setup will not provide any additional advantage over normal open farming conditions. LER > 1 would indicate a possible gain by using AV over PV-only or crop-only system.

Fig. 8 shows the mutual benefit (energy and crop) from single land use for different PV setups and panel density. By the definition in the above mentioned equation, the plots of Fig. 8 are essentially the sum of the energy ratio (Fig. 5) and crop ratio (Fig. 6). The energy ratio (Fig. 5) decreases with pitch p while the crop ratio increases (Fig. 6). Depending on the relative values of these ratios, in Fig. 8, we may get a possible optimum LER (e.g., An Giang, Jiangsu) or a decreasing LER (e.g., Rio Grande, Damietta).

There are cases when LER > 1, and AV seem viable. For example, in An Giang, the highest LER is 1.2 at $p = 4$ m for v-E/W configuration and the minimum value is 0.9 at $p = 2$ m for N/S AV farm. In Dhaka, both the tilted N/S and horizontal E/W compartments can achieve an additional 10% productivity. In Damietta and the Rio Grande, h-E/W provides an additional 45% and 55% production, respectively.

Finally, selling prices for the produced energy and paddy rice are not similar. Energy and crop yields should be weighted accordingly to find the true economic performance indicator as shown in Eqs. (15) and (16) (in Appendix A) – the AV optimization based on LER will not be appropriate. We will discuss AV optimization from an economic aspect in the next section. This will allow us to understand the impact of policy-level constraints on crop production.

5.2. *Economic output*

The annual revenue of the AV is calculated based on the revenues from energy and crop, as shown in Eq. 15 (in Appendix A). The energy revenue is shown in Fig. S2(a) follows the trend in energy yield shown earlier in Fig. 5. The same is of course true for the crop revenue (Fig. S2 (b)) and yield (Fig. 6). Note that, we simulate crop yield for a single season and then double that revenue to estimate the approximate two-season rice cropping revenue. Typically, the feed-in tariff (FIT) of

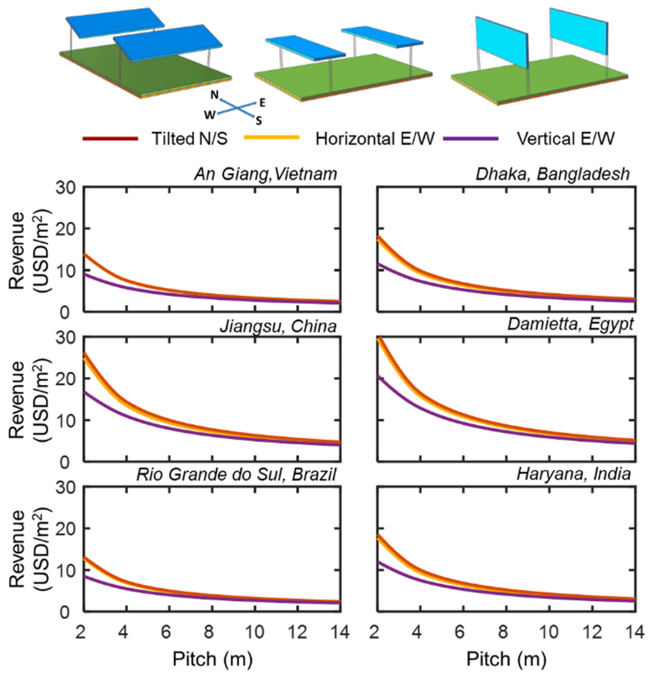

Fig. 9. Revenue as a function of the pitch for different AV configurations namely tilted N/S (Red), horizontal E/W (Orange), and vertical E/W (Purple) for different locations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

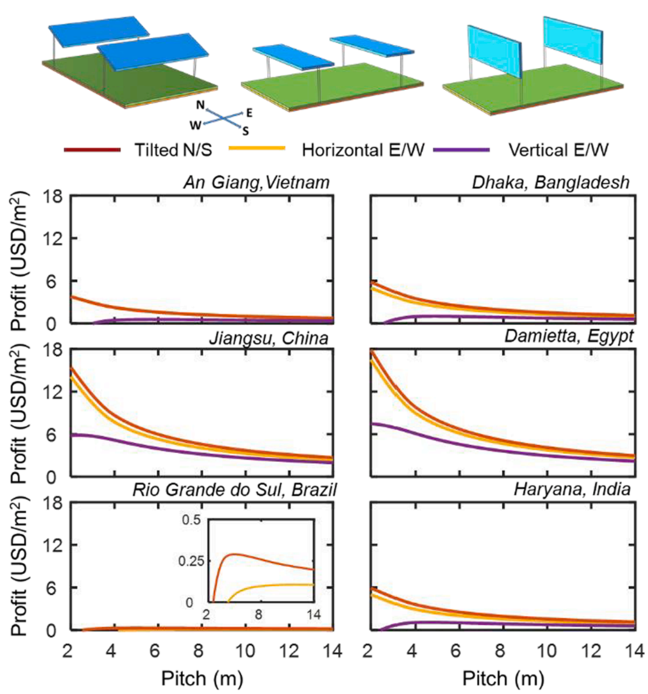

Fig. 10. Profit as a function of the pitch for different AV configurations namely tilted N/S (Red), horizontal E/W (Orange), and vertical E/W (Purple) for different locations. Inset shows a zoomed-in view of the profit profile for Rio Grande do Sul. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

electricity is much higher than the rice selling price – this is clearly observed in the energy and crop revenues in Figs. S2 (a) and (b). The high selling price and revenue (per land area) of energy will thus dominate the overall AV revenue. That is why the overall revenue of the AV system shown in Fig. 9 follows a trend versus pitch similar to that of the energy revenue. Both paddy production and PV energy output depend on local weather conditions. Although Haryana and Jiangsu produce a comparable amount of annual energy, we find a higher amount of revenue in China due to the higher feed-in tariff (FIT), see Table-S2. The lowest revenue is found in the Rio Grande do Sul (Brazil) – again, due to low FIT. Overall, the AV revenue shown in Fig. 9 will decrease with increasing pitch. The tilted N/S and h-E/W AV farms provide around similar revenue. Although the v-E/W AV configuration has higher crop production than the other two AV setups, we find lower revenue from it due to its lower energy contributions.

The annual profit margins from per unit land area for various bifacial AV setups, panel density, and locations are presented in Fig. 10. The highest profit is observed in Damietta (Egypt) and Jiangsu (China). This is because of the higher feed-in-tariff (FIT, see Table-S2) and the lower LCOE (see Eq. 16) in these locations. Paddy rice yields are also relatively high there. Apart from v-E/W, all AV arrangements provide a higher profit with decreasing pitch. At lower pitch values, mutual shading between vertically orientated arrays reduces energy contribution, which significantly increases LCOE (see Fig.S3) thereby lowering the AV profit. In some cases, the LCOE exceeds the solar energy selling price for the v-E/W setup. Then the AV system will not be economically viable compared to conventional farming.

Since FIT, banking rates and paddy producer prices are all influenced by governmental policy, in large-scale AV implementation, this policy is crucial. The policymakers of those countries may also want to set a minimum allowable reduction (10% or 20%) of crop production for AV systems. In such cases, AV would need to be designed to maximize profit while maintaining crop sustainability for the locality or country.

For a policy level constraint with a minimum crop ratio requirement of 90% or 80%, we would need to find the AV configuration and pitch for the maximum possible net profit. Towards such design, from Fig. 6, we find the pitch (p_{th}) of the AV for 90% (or 80%) crop ratio threshold. The optimal pitch will need to be at some value higher than p_{th} while maximizing the profit from Fig. 10. Typically, the optimal is the same as p_{th} due to the monotonically decreasing profit (Fig. 10). The optimal pitch design depends on the AV setup and the location.

Fig. 11(a) and Fig. 11(b) illustrate a comparison of optimal profit from various bifacial AV farms and traditional agricultural farms by assuring 90% and 80% paddy output, respectively. Note that our calculation assumes: two-season rice-cropping (one season fallow) and year-round energy generation. The optimal pitch values for all these designs are summarized in Fig. S4. In Fig. 11, for all locations, h-E/W AVs have the highest or close to highest profit. The v-E/W AV configuration provides lower profit for all locations. Assuring a 90% paddy output (see Fig. 11(a)), we find that the profit of AV farms can reach up to 22–115 times higher than conventional agricultural farming depending on the location. For An Giang, the h-E/W configurations provide a higher profit gain (around 30 times of conventional farming) than the other two AV setups. Although optimum pitch values for tilted N/S and h-E/W are the same in Dhaka (see Fig. S4), we found 4.4 times more profit gain from the tilted N/S AV setup due to higher energy contribution and lower LCOE (see Fig. 5 and Fig. S3). In Jiangsu, the highest profit can be found from the h-E/W setup (around 47 times conventional farming). However, tilted N/S and v-E/W AV setups provide around 35 and 36 times profit gains respectively. For Damietta, higher feed-in-tariff (see Table-S2) and higher insolation (see Fig. 2) lead to profit of around 115 times more than conventional farming. In Rio Grande do Sul, the economic gain of the h-E/W and tilted N/S AV farms are 1.2 and 3.2 times higher, respectively. Due to lower feed-in tariffs (FIT) and higher energy production costs (LCOE), establishing a v-E/W AV farm in the Rio Grande do Sul will not be economically viable. For Haryana, h-E/W performs best and provides around 65 times higher profit. Furthermore, if the policy level of minimum crop ratio is set at 80%, an extra 7%, 50%, 35%, 18%, and 26% profit may be obtained in An Giang, Dhaka, Jiangsu, Damietta, and Haryana, respectively (see

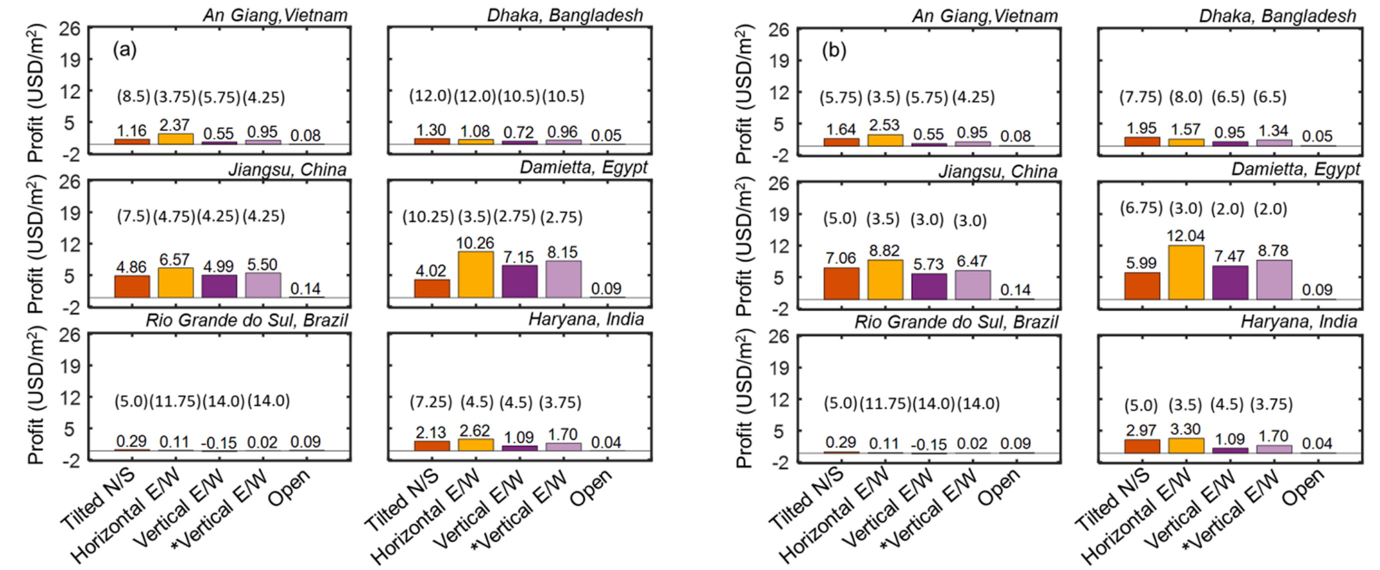

Fig. 11. Optimized profit for different bifacial AV farm configurations while ensuring (a) 90% and (b) 80% paddy production compared to conventional agricultural farms. The number within the bracket reflects the location-specific optimal pitch (in meters) for each AV arrangement. Note that, *Vertical E/W assumes a lower installation cost (120% of conventional PV) [38] compared to the installation costs (160% of conventional) of the other AV configurations considered. See Appendix A.3 for more details.

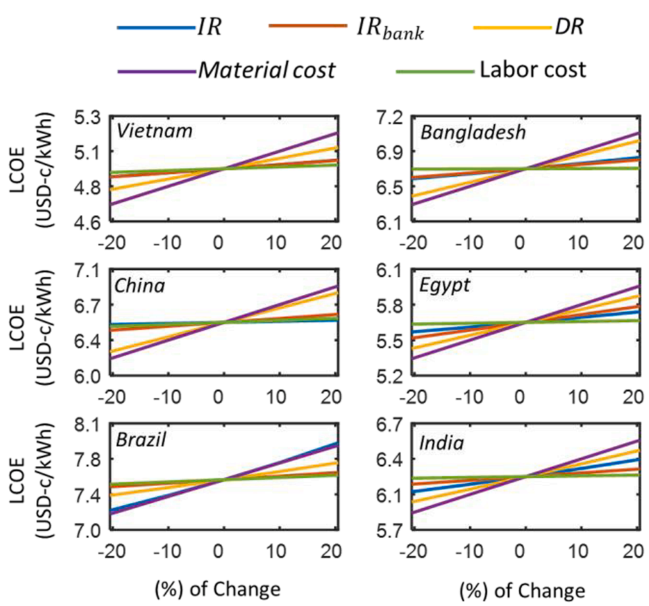

Fig. 12. Sensitivity analysis of LCOE in response to variations in the different banking rates, material cost of the system, and labor cost.

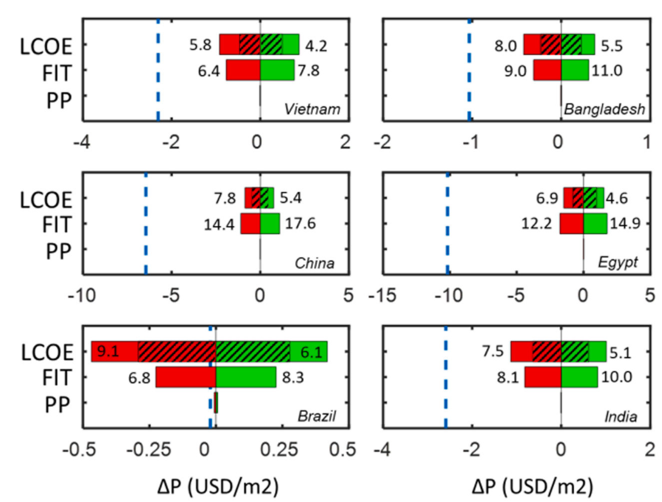

Fig. 13. Change in AV profit (ΔP) due to variation in LCOE, FIT and rice's producer's price (PP) is shown for horizontal E/W AV. The vertical dashed line represents the limit when the profit from AV will become lower than that of rice production. The limiting values of LCOE (USD-c/kWh) and FIT (USD-c/kWh) are mentioned within the plots. The hatched region indicates the scenario when the variation in material cost is not considered.

Fig. 11(b)). However, the profit margin from AV does not change even with the additional 10% reduction in crop production in Rio Grande do Sul. In terms of profit optimization, it is apparent from these results that the h-E/W-oriented bifacial AV farm would be more profitable for higher FIT offered countries. Although profits from N/S AV farms are higher in some cases, h-E/W should be the better choice due to the spatial uniformity of crop production.

For all our analysis, we have assumed a 60% higher installation cost for AV system compared to ground mounted PV array [22]. In practice, it may be possible to design vertical panel fixtures with costs 20% above conventional PV mountings [38]. Assuming such installation costs, the profits from vertical AV system will improve (due to lowered LCOE). This is marked as *Vertical E/W in Fig. 11.

The optimum design as well as the economic output from a monofacial AV farm would be different. Due to the large row spacing design in AVs, the bifacial panels have a significant advantage of albedo light collection. Even with a 7% higher installation cost for bifacial panels compared to monofacials, the bifacial PV provides enough additional energy for an 18 to 35% profit gain compared to the monofacial AV setup (See Fig. S5).

5.3. Sensitivity analysis

Various banking parameters such as the interest rate (IR_{bank}), inflation rate (IR), discount rate (DR) may vary depending on local economy. Material and labor costs for PV-system installation may change depending on local and global economy as well. Finally, feed-in-tariff and rice's selling price depends on local government policies. We, therefore, perform a sensitivity analysis to understand the impact of variations in economic parameters on the profit from AV systems. As discussed in the previous section, horizontal E/W (h-E/W) is the preferable option for economic gain and practicality. Thus, we focus on the h-E/W AV configuration in this section.

Fig. 12 presents the LCOE for varying the banking rates and initial investment by $\pm 20\%$ from the nominal value considered in the previous section. Within this range, LCOE shows a linear change—each of the lines indicate dependency of LCOE on a single parameter while other parameters are kept unchanged. LCOE has the strongest dependency on installation (material) cost. Among the banking rates, DR affects LCOE the most. The LCOE values at the limiting cases can be estimated where all the parameters simultaneously vary by $+20\%$ (or -20%). We use these limiting values of LCOE for the sensitivity analysis of profit in Fig. 13. The limiting LCOE values are also marked in this figure.

The profit (P) from the horizontal E/W AV (P_{AV}) and from rice in open field (P_{Rice}) have been discussed in Fig. 11. In Fig. 13, we find the change in AV profit (ΔP) due to variation in LCOE, feed-in-tariff (FIT) and rice's producer's price (PP). Here, LCOE ranges between the limits found from Fig. 12, and FIT and PP has $\pm 10\%$ and $\pm 20\%$ variations, respectively. The vertical dashed lines in Fig. 13 represent the threshold ΔP_{th} below which (i.e., for $\Delta P < \Delta P_{th}$) the AV becomes less profitable than rice production. All the three parameters: LCOE, FIT, and PP have approximately linear relationship with ΔP. In case more than one of these parameters have variations, the ΔP values will be summed.

We observe from Fig. 13 that, even with the limiting variations in LCOE and feed-in tariff (FIT), the AV remains highly profitable compared to rice production for Vietnam, Bangladesh, China, Egypt, and India. Feed-in tariff in Brazil was very close to LCOE, therefore, slight increase in LCOE or decrease in FIT makes AV (or any PV system) non-profitable. The effect of change in PP is very small.

Finally, one should realize that FIT for PV is often based on the policy decided at the time of the PV-system installation. If there is an increase in material cost or in banking rates, it may be likely that there will be rise in FIT as well. And, once the AV system is installed, the material cost becomes irrelevant in the LCOE sensitivity analysis discussed in Fig. 12. In such a case, the variation in ΔP is reduced as marked by the hatched regions in Fig. 13.

The sensitivity analysis presented here has been described in terms of fluctuations in costs and banking rates. The analysis also encompasses the situations where there are small errors or unpredictability in cost estimations. With the scaling of AV deployment and advancement in designs, we can expect reduction in installation costs and associated increase in profits. Overall, AV is likely to be highly profitable even after considering fluctuations in the economic factors.

6. Summary and conclusions

In this paper, we have presented an end-to-end, three stage techno-economic modeling framework to analyze the AV system's food-energy production efficacy and economic viability. *First*, we have used a self-consistent PV farm model to estimate the energy output and incident light on the crop surface of an AV farm. *Second,* we couple the irradiance information to a crop model of APSIM to estimate rice yield considering the non-uniform light distribution under the PV array. *Third,* the overall energy and rice production values are then used to calculate the revenue and profit using our economic model for AV farms.

Using our developed framework, we analyzed the AV system for six different rice-producing locations across the globe to evaluate the weather-aware performance and the respective economic viability. The key technical findings of this work are summarized below:

1. The food-energy production of AV farms strongly depends on the local ambient conditions. PV energy output increases with insolation (see Fig. 2 and Fig. 3). Rice yield, on the other hand, may not change significantly at high insolation values (see Appendix A.2). Implementing an AV farm at locations of higher irradiance such as Damietta (Egypt) and Haryana (India) will provide additional benefits by providing higher energy output.
2. Using our modeling framework, we have investigated the PV energy and rice yield variations for three different AV farm configurations (tilted N/S, horizontal E/W, and vertical E/W). The energy output of the horizontal E/W and tilted N/S AV farm is similar, while the vertical array yields the lowest (see Fig. 5). The rice yield under the PV panel array of an AV farm depends on the light distribution. For vertical AV setup, light distribution on the crop surface and the rice production is higher than the other two configurations due to the array orientation and highly tilted PV arrangement (see Fig. 1(c)).
3. As the array pitch (or the p/h ratio) increases, the panel density over a given land decreases, thereby reducing the energy yield per land area. The higher panel spacing also enhances the light collected on the crop surface, improving rice production. The PV energy therefore non-linearly decreases with increasing pitch and then saturates at higher array pitch values. Rice production on the other hand increases with pitch and then saturates.

While these technical findings indicate an optimum orientation and pitch for combined food-energy production, one should consider economic factors to evaluate AV performance. Our relevant findings on AV economics are as follows:

1. Several prior works have used land equivalent ratio, LER (i.e., the normalized net output of food and energy) as the performance indicator for an AV system. As discussed in this paper, the economic weights of energy and rice are significantly different, and LER may be an inadequate measure of AV performance.
2. Economic output (revenue and profit) from an AV configuration is dependent on local economics and the associated renewable energy policies. Feed-in-tariff and additional banking rates play an important role in making the AV system economically viable. Considering the country-specific economic parameters, we found around 22 to 115 times more profit than conventional paddy rice farming (see Fig. 11(a) and Fig. 11(b)) among the six locations in our study.
3. A sensitivity analysis for horizontal E/W AV system shows that, except for Brazil, high profit gain from AV is expected even with 20% higher banking rates or material costs. We observe a low profit margin for AV (or PV) for Brazil with the current banking rates, costs, and feed-in-tariff (FIT). As a result, positive fluctuations in rates and costs may lead to a non-profitable AV system. However, the unusually small margin between LCOE and FIT in Brazil requires further

investigation. After all, FIT can regularly be updated based on recent economic conditions.

4. Due to higher fixtures and larger panel spacing, the bifacial gain is higher in AV farms compared to utility-scale PV arrays. Using bifacial PV instead of monofacial PV may add 18 to 35% extra profit (See Fig. S5) for AV systems.

Finally, because of the higher investment opportunity and net profit from PV, energy generation per unit land area has a greater impact on the economic output of an AV farm than rice production. Typically, the design towards a denser PV array (even with drastically reduced rice production) is economically more favorable in an AV system. This means there should be a policy-level, minimum rice production requirement for food security if AV configurations are to be adopted. For example, within the six locations considered, to meet a requirement of 90% rice production (compared to open field rice cultivation) in an AV system, the pitch of the PV array varies between 3.5 and 12 m for horizontal E/W. We predict the net profit to be 22–115 times higher in AVs (with 90% rice yield constraint) compared to only producing rice in those lands.

CRediT authorship contribution statement

M. Sojib Ahmed: Methodology, Software, Data curation, Visualization, Writing – original draft, Writing – review & editing. **M. Rezwan Khan:** Conceptualization, Supervision, Writing – review & editing. **Anisul Haque:** Conceptualization, Supervision, Writing – review & editing. **M. Ryyan Khan:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We gratefully thank feedback from Prof. Syed Basher (East West University, Bangladesh). The work was funded by UIU-IAR, ref. UIU/IAR/02/2019-20/SE/09. We also thank United International University (UIU) and Bangladesh Research and Education Network (BdREN) for providing computational support. We also thank the anonymous reviewers for the comments and suggestions to improve the work and the paper.

Appendix A. Modeling methods

A summary of the AV farm modeling framework has been discussed in Section 2. In the following subsections, we will discuss the details for the numerical modeling of PV-array, crop yield and economics.

A.1. Solar farm modeling:

There are several prior models for irradiance and PV farm analysis [32–34,39,40]. In this work, we will use the Purdue view factor model [34,41]. The equations for the front side of the panel are listed in Tables 1 and 2. The backside of the bifacial panel follows a similar approach. The models are briefly described here.

For irradiance modeling, we use the Sandia PVLib library [42] to calculate the sun's path and the corresponding time-varying solar angles (e.g., the azimuth γ_s and zenith angles θ_Z of the sun) for any given location. The time-varying irradiance data is obtained from the Haurwitz clear sky model [43,44]. We then normalize it using location-specific insolation data from NASA's POWER database [30] to estimate a more precise location-specific irradiance I_{GHI} (GHI: Global Horizontal Irradiance). Finally, I_{GHI} is then split into direct (I_{DNI}) and diffuse (I_{DHI}) light using Orgill-Hollands [45] and Perez models [46]. So, location information and GHI data are the main input parameter for irradiance modeling.

After obtaining time-varying direct and diffuse light, we use Eqs. 1–5 in Table 1 to calculate the direct $I_{PV:DNI}(l)$ and diffuse $I_{PV:DHI}(l)$ light at any position l on the front (F) face of the panel as well as the light distribution over the ground $I_{gnd}(x)$. These quantities depend on the solar farm configuration. Relevant view factors have been used to quantify the fractional sky-diffuse light on the ground and the panel faces [33,34,47]. For a

Table 1
Summary of equations for light collection model.

Light collection on the panel from the sky	
$I^F_{PV:DNI}(l) = \begin{cases} I_{DNI}\ cos\theta_F\ (1-R(\theta_F))\eta_{dir} & ; l \text{ is not shaded} \\ 0 & ; l \text{ is shaded} \end{cases}$	(1)
$I^F_{PV:DHI}(l) = I_{DHI} \times F_{dl-sky}(l) \times \eta_{diff}$	(2)
Light collection on the ground from the sky	
$I_{gnd:DNI}(x) = \begin{cases} I_{DNI}\ \cos\theta_Z & ;\ x \text{ is not shaded} \\ 0 & ;\ x \text{ is shaded} \end{cases}$	(3)
$I_{gnd:DHI}(x) = I_{DNI} \times F_{dx-sky}(x)$	(4)
$I_{gnd}(x) = I_{gnd:DNI}(x) + I_{gnd:DHI}(x)$	(5)
Light collection on the panel from the ground	
$I^F_{PV:Alb}(l) = \eta_{diff} \int_0^p R_A \times I_{gnd}(x) \times F_{dl-dx}(x,l)dx$	(6)
$I^F_{PV}(l) = I^F_{PV:DNI} + I^F_{PV:DHI} + I^F_{PV:Alb}$	(7)
$I_{PV}(l) = I^F_{PV}(l) + I^B_{PV}(l)$	(8)

Table 2
Thermal model.

Temperature-dependent PV efficiency model	
$T_{cell} = T_{amb} + c_T \frac{\tau\alpha}{u_L} P_{POA}\left(1 - \frac{\eta(T_{cell})}{\tau\alpha}\right)$	(9)
$c_T \approx 1$; $\tau = 0.95$; $\alpha = 0.8$; $u_L = 21.5$	
$\eta\ (T_{cell}) = \eta_{STC}(1 - TC(T_{cell} - T_{STC}))$	(10)
$P_{POA} = P_{PV:DNI}/\eta_{dir} + (P_{PV:DHI} + P_{PV:Alb})/\eta_{diff}$	(11)
$P_{PV:T} = \eta(T_{cell}) \times P_{POA}$	(12)

given albedo, ground reflectance contribution $I_{PV:Alb}(l)$ to the bifacial panel can be calculated using Eq. 6 – this requires the ground surface reflectivity value R_A. Studies [40,48] have shown that this albedo value of crop varies between 0.15 and 0.3. In our numerical calculations, we assumed the average ($R_A = 0.23$) value of this range. Finally, we get the total light collection of the panel front face by combining three contributing components (direct, diffuse, and albedo) for each time step (Eq. 7). The light collection on the back face can be similarly calculated and combined with that of the front to get the light collection on the panel (Eq. 8).

The light conversion efficiencies of the solar cell degrade with its temperature, which in turn is proportional to the ambient temperature T_{amb} and the plane of array irradiance P_{POA} (see Eq. 9 and Eq. 10). Although the bifacial module collects more irradiance than the monofacial, the efficiency degradation of the bifacial module is much lower than the monofacial due to the lower temperature coefficient (TC)[39]. In this work, we have used Skoplaski's model to calculate the cell temperature T_{Cell} and the efficiency $\eta(T_{Cell})$. For bifacial Si Hetero-junction ($\eta_{STC} = 20.9\%$, $TC = 0.26\%/^{\circ}C$) necessary coefficients and input parameters of Eq. 9 and Eq. 10 in Table 2 are collected from ref. [39].

We can now compute daily energy output by multiplying the adjusted PV efficiency by the daily integrated light collected by the panels. We assume three bypass diodes in a panel – our electrical model appropriately includes these to calculate the power output under shading and non-uniform illumination. Finally, we integrate the power to find the daily energy and then the yearly energy yield (YEY) per land area.

A.2. Rice yield modeling:

In our modeling framework, we couple our illumination model with APSIM-Oryza [28] to investigate rice growth and responses under PV arrays. Previous studies [49,50] by different groups have shown that the performance of this cropping model in simulating aboveground biomass and crop yield matches well with the experimental study. For the APSIM-Oryza simulation framework, the ORYZA2000 rice growth model [51] has been used to determine crop phonology, organ development, and response to abiotic stresses. This biophysical cropping model can estimate daily potential crop production for any given weather, soil, and management conditions. Several initializations for different modules are required to simulate the intercrop model, such as daily weather data (e.g., solar radiation, maximum temperature, minimum temperature, rainfall), soil characterization data (e.g., available water, organic carbon, nitrogen, and pH at different depths of the soil), and crop management data (e.g., duration of the seed sowing and ponding, number of plants per hill, number of hills, fertilizer type, maximum depth of the pond, minimum ponding depth to start irrigation, etc.). The potential crop output is then estimated according to radiation use efficiency (RUE) driven by weather conditions and available leaf area [49]. The actual production can be found after limiting the yield due to abiotic stresses. In our study, we have used soil and management information from the default APSIM simulation example for cultivating paddy rice in south Asia. We also consider automated irrigation to ensure enough water for crop growth instead of relying solely on rainfall.

We can get a good idea of rice yield response for combined variations in average temperature and daily radiation from Fig. 14. This depicts rice yield as a function of daily average radiation and temperature with considering automatic irrigation (with no rain). The effects of average temperature (maximum and minimum temperatures are $\pm 10^{\circ}$C of the average temperature) and radiation on paddy rice production are significant. We can see that rice production is best at temperatures ranging between 20 to 30°C. The yield sharply declines away from these temperatures. Additionally, rice production increases with daily average radiation and eventually starts to saturate beyond average radiation of 20 MJ/m^2/day. There are some noticeable oscillations in Fig. 14. These may be calibration issues arising from the numerical estimation of default sub-modules within APSIM. Typically, it is difficult to accurately predict crop production for a given year or season due to uncertainty in weather and crop-evolution—statistical

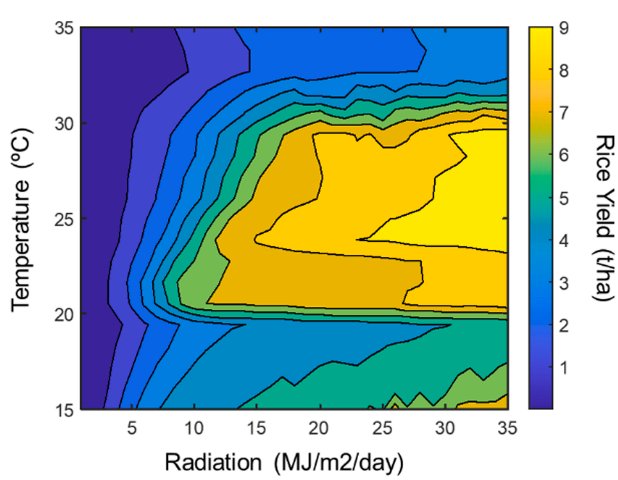

Fig. 14. APSIM [28] rice yield model response as a function of radiation and average temperature. The weather parameters (radiation, temperature) are considered constant over time and there is automatic irrigation (with no rain).

predictions, as done here, would be more consistent. The relative change in crop production under partial shading compared to open conditions is an important performance indicator in the AV system. Small mismatch in crop yield may negligibly affect our analysis. The parametric rice yield map shown in Fig. 14 helps us understand how much shadow (radiation reduction) is allowable for AV design and under which weather conditions we may be able to implement such systems.

For the AV setup, non-uniform radiation distribution on the cropland $I_{gnd}(t,x)$ (see Eq. 5 in the solar farm model) needs to be considered. Even for a given PV array configuration (e.g., fixture height, array direction, panel tilt), this non-uniform radiation distribution will depend on the day of the year, time of day, and location. The crop model in APSIM uses daily radiation energy as input. We, therefore, integrate the time-varying irradiance to find the spatially distributed daily radiation (*for a given day*) as follows:

$$I_{gnd:daily}(x) = \int_{Sunrise}^{Sunset} I_{gnd}(t,x)dt.$$

This radiation will then be used as input to APSIM to calculate rice growth and production at each point on the ground. $I_{gnd:daily}(x)$ will of course vary from day to day. Additionally, other location-specific information such as the ambient temperature, rainfall, soil, and crop management data needs to be coupled with rice yield simulations. Next, we calculate the spatially distributed rice yield for a particular crop season $CY_{S.Season}(x)$ – this result is obtained by assuming separate APSIM simulations at each grid-point on the ground. Finally, the average rice yield is calculated as follows:

$$CY_{Avg:S.Season} = \frac{1}{p}\int_0^p CY_{S.Season}(x)dx$$

In general, the number of rice-producing seasons determines the yearly crop yield (YCY). Many potential rice-producing countries cultivate rice twice a year [52]. For simplicity, we simulate rice production for a single season and assume that we will get double that amount yearly.

A.3. Economic modeling:

To analyze the economic output of an AV farm, first, we need to calculate the PV energy production cost. The Levelized Cost of Energy (LCOE) depends on the initial investment (own and bank loans), warranty and insurance, maintenance cost, and the yearly energy yield (YEY). Equations used to calculate LCOE are summarized in Table-S1 and Eq.13–14 in Table 3 [53]. Several economic parameters are required for LCOE calculations, such as banking rates (e.g., inflation rate IR, bank interest rate IR_{bank}, and discount rate DR) and labor cost data are location specific. All necessary data and relevant references are provided in the supplementary information; see Table-S2.

We use the LCOE-calculation approach similar to that described in ref. [2,53]. The AV system, however, requires a higher elevated PV setup than the conventional ground-mounted PV to operate agricultural activity, which increases the installation cost by around 60% for reinforcement mount AV setup [22,38] which translates to material cost of 0.45 USD/W_P – see section S1 (supp. doc.) for the calculation breakdown. We consider this amount constant for all configurations in our study. In practice, however, tilted and horizontal panel arrays would need the reinforced structures while vertical panels may be supported by a less-fortified structures. Hence, the material cost of vertical AV installation can be estimated to be 20% higher than ground mounted PV (i.e., 0.34 USD/W_P) [38].

Additionally, we assumed that 60% of the initial investment comes from a bank loan with a 10-year tenor, and the rest comes from own investment C_{own} [2]. There will be interest $C_{bank,int}$ on borrowing amount along with amortization $C_{bank,amor}$. The insurance C_{insu}, inverter warranty cost C_{war}, and operation and maintenance cost C_{OM} are the parts of the total system cost. Total system cost C_{PV} (USD/W_P) is then calculated using Eq. 13 in Table 3 with considering 25 years system lifetime I_s. The breakdown of each terms of the equation is provided in the supplementary information; see Tabe-S1. For calculating LCOE using Eq. 14, we can see that along with the PV system cost and the yearly energy yield (YEY, described in PV system modeling – Appendix A.1) over the lifetime of the system is used in Eq. 14. For this study, we consider the degradation rate to be 1%/year [54]. The yearly energy yield in Eq. 14 is correspondingly discounted to calculate LCOE.

We calculate the gross yearly revenue (YR) in Eq. 15 as the sum of PV energy and paddy rice based on the local feed-in-tariff (FIT) and the producer price (PP), respectively. A recent study [55] shows that paddy cultivation may provide around $\sim$ 20% profit over the total revenue. For simplicity, in this study we assumed paddy rice profit ratio $PR = 0.20$ for all the locations under study – this means, rice production profit is $PR\times$ revenue. The profit from PV is calculated by subtracting PV energy generation cost (LCOE $\times$ YEY) from its revenue (FIT $\times$ YEY). The net yearly profit (YP) is the sum of the profits from crop and PV – the profit calculations are summarized in Eq. 16. (See Table 4).

Table 3
Economic model.

$C_{PV} = C_{bank,int} + C_{bank,amor} + C_{own} + C_{war} + C_{insu} + C_{OM}$	(13)
$\mathrm{LCOE_{AV}} = \dfrac{C_{PV}}{\sum_{i=0}^{i=I_s}\dfrac{YEY_{AV}(i)}{(1+DR)^i}}$	(14)
$YR_{AV} = \mathrm{FIT} \times YEY_{AV} + \mathrm{PP} \times YCY_{AV}$	(15)
$YP_{AV} = (\mathrm{FIT} - \mathrm{LCOE_{AV}}) \times YEY_{AV} + PR \times PP \times YCY_{AV}$	(16)

Table 4
Symbols used in this paper.

Parameters	Definition
TC	Temperature Coefficient (%/°C)
T_{amb}	Ambient Temperature (°C)
T_M	Module Temperature (°C)
η	Power conversion efficiency (%)
c_T	Correction term used for the daily average temperature data
τ	Transmittance of glazing (no unit)
α	Absorbed fraction (no unit)
u_L	Heat loss coefficient (W/m^2K)
θ_Z	Zenith Angle (*degrees*)
θ_F	Angle of incidence at the front face of the panel
Pitch (p)	Row-to-row distance between the bottom edges of consecutive arrays (m)
Height (h)	Height of the panel
y_e	Elevation
β	Tilt angle
γ_A	Azimuth angle from measure from the North
R_A	Albedo
l_s	Lifetime of a farm (in years)
YEY_{AV}	Yearly Energy Yield ($kWh/m^2/yr$)
YCY_{AV}	Yearly Crop (paddy) Yield ($kg/m^2/yr$)
C_{PV}	Total system cost (USD/W_P)
$P_{STC,f}$	Direct power production from the front face of the PV at standard test conditions (W_P)
IR_{bank}	Bank interest rate (%)
IR	Inflation rate (%)
dt_{bank}	Bank tenor (yr.)
DR	Discount rate (%)
l_b	(%) of total system cost comes from bank lone
c_{insu}	(%) of initial investment for insurance
C_{SP}	Solar panel cost (USD/W_P)
C_{inv}	Inverter cost (USD/W_P)
W_{inv}	Inverter warranty (yr.)
π_{inv}	Inverter warranty extension cost
$c_{OM,mat}$	Material cost for operation and maintenance ($USD/W_P/yr.$)
PP	Producer price of paddy rice (USD/t)
FIT	Feed-in tariff (USD/kWh)

Sub/Super-scripts	Definition
DHI	Diffuse Horizontal Irradiance
diff	Diffuse
dir	Direct
DNI	Direct Normal Irradiance
F	Front
Farm	PV Farm
GHI	Global Horizontal Irradiance
AV	Agrivoltaics
O&M	Operation and maintenance
Panel	PV panel
PV	Photovoltaic
STC	Standard testing condition

Appendix B. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2022.119560.

References

[1] Statistical Review of World Energy | Energy economics | Home. bp global. https://www.bp.com/en/global/corporate/energy-economics/statistical-review-of-world-energy.html (accessed Sep. 07, 2021).

[2] Rodríguez-Gallegos CD, Liu H, Gandhi O, Singh JP, Krishnamurthy V, Kumar A, et al. Global techno-economic performance of bifacial and tracking photovoltaic systems. Joule 2020;4(7):1514–41. https://doi.org/10.1016/j.joule.2020.05.005.

[3] International Technology Roadmap for Photovoltaic (ITRPV), 12th Editions. VDMA (2021). Accessed: Sep. 07, 2021. [Online]. Available: https://itrpv.vdma.org/en/.

[4] Expert meeting on How to Feed the World in 2050. FAO headquarters, 24-26 June 2009, FOA. https://www.fao.org/3/ak542e/ak542e00.htm (accessed Jan. 20, 2022).

[5] Miskin CK, Li Y, Perna A, Ellis RG, Grubbs EK, Bermel P, et al. Sustainable co-production of food and solar power to relax land-use constraints. Nat. Sustain. 2019;2(10):972–80.

[6] Lee K, Khanal S, Bakshi BR. Techno-ecologically synergistic food–energy–water systems can meet human and ecosystem needs. Energy Environ. Sci. 2021;14(7):3700–16. https://doi.org/10.1039/D1EE00843A.

[7] Marrou H, Guilioni L, Dufour L, Dupraz C, Wery J. Microclimate under agrivoltaic systems: is crop growth rate affected in the partial shade of solar panels? Agric. For. Meteorol. 2013;177:117–32. https://doi.org/10.1016/j.agrformet.2013.04.012.

[8] Marrou H, Wery J, Dufour L, Dupraz C. Productivity and radiation use efficiency of lettuces grown in the partial shade of photovoltaic panels. Eur. J. Agron. 2013;44:54–66. https://doi.org/10.1016/j.eja.2012.08.003.

[9] Hassanpour Adeh E, Selker JS, Higgins CW, Villarini M. Remarkable agrivoltaic influence on soil moisture, micrometeorology and water-use efficiency. PLoS ONE 2018;13(11):e0203256. https://doi.org/10.1371/journal.pone.0203256.

[10] Barron-Gafford GA, Pavao-Zuckerman MA, Minor RL, Sutter LF, Barnett-Moreno I, Blackett DT, et al. Agrivoltaics provide mutual benefits across the food–energy–water nexus in drylands. Nat Sustain 2019;2(9):848–55.

[11] Weselek A, Bauerle A, Zikeli S, Lewandowski I, Högy P. Effects on crop development, yields and chemical composition of celeriac (*Apium graveolens* L. var. rapaceum) cultivated underneath an agrivoltaic system. Agronomy 2021;11(4):733. https://doi.org/10.3390/agronomy11040733.

[12] Valle B, Simonneau T, Sourd F, Pechier P, Hamard P, Frisson T, et al. Increasing the total productivity of a land by combining mobile photovoltaic panels and food crops. Appl Energy 2017;206:1495–507. https://doi.org/10.1016/j.apenergy.2017.09.113.

[13] Peng S, Huang J, Sheehy JE, Laza RC, Visperas RM, Zhong X, et al. Rice yields decline with higher night temperature from global warming. Proc Natl Acad Sci 2004;101(27):9971–5. https://doi.org/10.1073/pnas.0403720101.

[14] Gonocruz RA, Nakamura R, Yoshino K, Homma M, Doi T, Yoshida Y, et al. Analysis of the rice yield under an agrivoltaic system: a case study in Japan. Environments 2021;8(7):65. https://doi.org/10.3390/environments8070065.

[15] Hatfield JL, Prueger JH. Temperature extremes: Effect on plant growth and development. Weather Clim Extrem 2015;10:4–10. https://doi.org/10.1016/j.wace.2015.08.001.

[16] Ammapet Vijayan R, Sivanarul J, Varadharajaperumal M. Optimizing the spectral sharing in a vertical bifacial agrivoltaics farm. J Phys Appl Phys 2021;54(30):304004. https://doi.org/10.1088/1361-6463/abfbae.

[17] Dupraz C, Marrou H, Talbot G, Dufour L, Nogier A, Ferard Y. Combining solar photovoltaic panels and food crops for optimising land use: towards new agrivoltaic schemes. Renew Energy 2011;36(10):2725–32. https://doi.org/10.1016/j.renene.2011.03.005.

[18] Trommsdorff M, Kang J, Reise C, Schindele S, Bopp G, Ehmann A, et al. Combining food and energy production: Design of an agrivoltaic system applied in arable and vegetable farming in Germany. Renew Sustain Energy Rev 2021;140:110694. https://doi.org/10.1016/j.rser.2020.110694.

[19] Riaz MH, Imran H, Younas R, Alam MA, Butt NZ. Module technology for agrivoltaics: vertical bifacial versus tilted monofacial farms. IEEE J Photovolt 2021;11(2):469–77. https://doi.org/10.1109/JPHOTOV.2020.3048225.

[20] Riaz MH, Imran H, Alam H, Alam MA, Butt NZ. Crop-specific Optimization of Bifacial PV Arrays for Agrivoltaic Food-Energy Production: The Light-Productivity-Factor Approach. ArXiv210400560 Phys., Apr. 2021, Accessed: Aug. 28, 2021. [Online]. Available: http://arxiv.org/abs/2104.00560.

[21] Campana PE, Stridh B, Amaducci S, Colauzzi M. Optimization of vertically mounted agrivoltaic systems. ArXiv210402124 Cs Eess, Apr. 2021, Accessed: Apr. 16, 2021. [Online]. Available: http://arxiv.org/abs/2104.02124.

[22] Schindele S, Trommsdorff M, Schlaak A, Obergfell T, Bopp G, Reise C, et al. Implementation of agrophotovoltaics: Techno-economic analysis of the price-performance ratio and its policy implications. Appl Energy 2020;265:114737. https://doi.org/10.1016/j.apenergy.2020.114737.

[23] Thompson EP, Bombelli EL, Shubham S, Watson H, Everard A, D'Ardes V, et al. Tinted semi-transparent solar panels allow concurrent production of crops and electricity on the same cropland. Adv Energy Mater 2020;10(35):2001189. https://doi.org/10.1002/aenm.202001189.

[24] Dinesh H, Pearce JM. The potential of agrivoltaic systems. Renew Sustain Energy Rev 2016;54:299–308. https://doi.org/10.1016/j.rser.2015.10.024.

[25] Feuerbacher A, Laub M, Högy P, Lippert C, Pataczek L, Schindele S, et al. An analytical framework to estimate the economics and adoption potential of dual land-use systems: The case of agrivoltaics. Agric Syst 2021;192:103193. https://doi.org/10.1016/j.agsy.2021.103193.

[26] Amaducci S, Yin X, Colauzzi M. Agrivoltaic systems to optimise land use for electric energy production. Appl Energy 2018;220:545–61. https://doi.org/10.1016/j.apenergy.2018.03.081.

[27] "World's Leading Rice Producing Countries," *AtlasBig.* https://www.atlasbig.com/en-ae/countries-by-rice-production (accessed Oct. 20, 2021).

[28] Holzworth DP, Huth NI, deVoil PG, Zurcher EJ, Herrmann NI, McLean G, et al. APSIM – evolution towards a new generation of agricultural systems simulation. Environ Model Softw 2014;62:327–50. https://doi.org/10.1016/j.envsoft.2014.07.009.

[29] Keating BA, Carberry PS, Hammer GL, Probert ME, Robertson MJ, Holzworth D, et al. An overview of APSIM, a model designed for farming systems simulation. Eur J Agron 2003;18(3-4):267–88. https://doi.org/10.1016/S1161-0301(02)00108-9.

[30] NASA Prediction of Worldwide Energy Resources (POWER) Data Access Viewer. https://power.larc.nasa.gov/data-access-viewer/ (accessed Jul. 26, 2021).

[31] FAO GIEWS Country Briefs-Home. https://www.fao.org/giews/countrybrief/index.jsp (accessed Dec. 22, 2021).

[32] Sun X, Khan MR, Deline C, Alam MA. Optimization and performance of bifacial solar modules: a global perspective. Appl Energy 2018;212:1601–10. https://doi.org/10.1016/j.apenergy.2017.12.041.

[33] Khan MR, Hanna A, Sun X, Alam MA. Vertical bifacial solar farms: physics, design, and global optimization. Appl Energy 2017;206:240–8. https://doi.org/10.1016/j.apenergy.2017.08.042.

[34] Patel MT, Khan MR, Sun X, Alam MA. A worldwide cost-based design and optimization of tilted bifacial solar farms. Appl Energy 2019;247:467–79. https://doi.org/10.1016/j.apenergy.2019.03.150.

[35] Information I-WS. SoilGrids web portal. SoilGrids web portal. https://soilgrids.org (accessed Jan. 20, 2022).

[36] Subash N, Shamim M, Singh VK, Gangwar B, Singh B, Gaydon DS, et al. Applicability of APSIM to capture the effectiveness of irrigation management decisions in rice-based cropping sequence in the Upper-Gangetic Plains of India. Paddy Water Environ 2015;13(4):325–35. https://doi.org/10.1007/s10333-014-0443-1.

[37] FAOSTAT. http://www.fao.org/faostat/en/#data/QC (accessed Apr. 21, 2021).

[38] Horowitz K, Ramasamy V, Macknick J, Margolis R. Capital costs for dual-use photovoltaic installations: 2020 benchmark for ground-mounted PV systems with pollinator-friendly vegetation, grazing, and crops. NREL/TP–6A20-77811, 1756713, MainId:30726, December 2020. doi: 10.2172/1756713.

[39] Patel MT, Vijayan RA, Asadpour R, Varadharajaperumal M, Khan MR, Alam MA. Temperature-dependent energy gain of bifacial PV farms: a global perspective. Appl Energy 2020;276:115405. https://doi.org/10.1016/j.apenergy.2020.115405.

[40] Patel MT, Ahmed MS, Imran H, Butt NZ, Khan MR, Alam MA. Global analysis of next-generation utility-scale PV: tracking bifacial solar farms. Appl Energy 2021;290:116478. https://doi.org/10.1016/j.apenergy.2021.116478.

[41] Khan MR, Patel MT, Asadpour R, Imran H, Butt NZ, Alam MA. A review of next generation bifacial solar farms: predictive modeling of energy yield, economics, and reliability. J Phys D: Appl Phys 2021;54(32):323001. https://doi.org/10.1088/1361-6463/abfce5.

[42] Stein JS, Holmgren WF, Forbess J, Hansen CW. PVLIB: Open source photovoltaic performance modeling functions for Matlab and Python. In: 2016 IEEE 43rd Photovoltaic Specialists Conference (PVSC), Jun. 2016, pp. 3425–3430. doi: 10.1109/PVSC.2016.7750303.

[43] Haurwitz B. Insolation in relation to cloud type. J Atmospheric Sci 1946;3(4):123–4. https://doi.org/10.1175/1520-0469(1946)003<0123:IIRTCT>2.0.CO;2.

[44] Neumann J. Insolation in relation to cloud amount. Mon Weather Rev 1954;82(11):317–9. https://doi.org/10.1175/1520-0493(1954)082<0317:IIRTCA>2.0.CO;2.

[45] Orgill JF, Hollands KGT. Correlation equation for hourly diffuse radiation on a horizontal surface. Sol Energy 1977;19(4):357–9. https://doi.org/10.1016/0038-092X(77)90006-8.

[46] Perez R, Seals R, Ineichen P, Stewart R, Menicucci D. A new simplified version of the perez diffuse irradiance model for tilted surfaces. Sol Energy 1987;39(3):221–31. https://doi.org/10.1016/S0038-092X(87)80031-2.

[47] Radiative Heat Transfer. Elsevier; 2013. doi: 10.1016/C2010-0-65874-3.

[48] Carrer D, Pique G, Ferlicoq M, Ceamanos X, Ceschia E. What is the potential of cropland albedo management in the fight against global warming? A case study based on the use of cover crops. Environ Res Lett 2018;13(4):044030. https://doi.org/10.1088/1748-9326/aab650.

[49] Gaydon DS, Balwinder-Singh, Wang E, Poulton PL, Ahmad B, Ahmed F, et al. Evaluation of the APSIM model in cropping systems of Asia. Field Crops Res 2017;204:52–75. https://doi.org/10.1016/j.fcr.2016.12.015.

[50] Gaydon DS, Probert ME, Buresh RJ, Meinke H, Suriadi A, Dobermann A, et al. Rice in cropping systems—Modelling transitions between flooded and non-flooded soil environments. Eur J Agron 2012;39:9–24. https://doi.org/10.1016/j.eja.2012.01.003.

[51] Bouman BAM, van Laar HH. Description and evaluation of the rice growth model ORYZA2000 under nitrogen-limited conditions. Agric Syst 2006;87(3):249–73. https://doi.org/10.1016/j.agsy.2004.09.011.

[52] Laborte AG, Gutierrez MA, Balanza JG, Saito K, Zwart SJ, Boschetti M, et al. RiceAtlas, a spatial database of global rice calendars and production. Sci Data 2017;4(1). https://doi.org/10.1038/sdata.2017.74.

[53] Rodríguez-Gallegos CD, Bieri M, Gandhi O, Singh JP, Reindl T, Panda SK. Monofacial vs bifacial Si-based PV modules: Which one is more cost-effective? Sol Energy 2018;176:412–38. https://doi.org/10.1016/j.solener.2018.10.012.

[54] Jordan DC, Kurtz SR, VanSant K, Newmiller J. Compendium of photovoltaic degradation rates. Prog Photovolt Res Appl 2016;24(7):978–89. https://doi.org/10.1002/pip.2744.

[55] Suwanmaneepong S, Kerdsriserm C, Lepcha N, Cavite HJ, Llones CA. Cost and return analysis of organic and conventional rice production in Chachoengsao Province, Thailand. Org Agric 2020;10(3):369–78. https://doi.org/10.1007/s13165-020-00280-9.